

BOULDER, CO, USA

WMO: 720533

Lat: 40.033N

Lon: 105.217W

Elev: 5289

StdP: 12.10

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 00160

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	4.6	9.6	-6.1	4.8	27.6	-1.8	6.1	30.6	32.3	51.5	26.3	47.4	2.1	50	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	94.8	59.4	90.9	59.0	89.5	59.0	64.5	78.4	63.4	78.5	62.1	77.6	6.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	98.5	66.8	59.3	91.9	66.7	57.2	85.2	67.3	32.6	77.4	31.6	77.9	30.6	77.7	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.1	18.6	15.1	DB	-3.6	99.2	5.9	2.0	-7.8	100.6	-11.3	101.8	-14.6	103.0	-18.9	104.4
			WB	-3.4	67.1	6.1	1.7	-7.8	68.3	-11.4	69.3	-14.8	70.3	-19.2	71.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	35.1	34.6	44.5	49.6	55.9	69.9	74.6	73.2	66.2	51.9	42.3	33.8
	DBStd	17.03	9.77	11.63	10.06	8.41	9.07	6.58	4.34	4.85	7.27	9.68	9.70	10.33
	HDD50	2153	467	443	222	107	51	0	0	0	3	92	260	508
	HDD65	5342	928	852	635	463	298	26	1	7	72	409	683	968
	CDD50	3149	4	11	53	95	235	598	762	718	488	152	28	5
	CDD65	861	0	0	1	1	16	174	298	260	107	4	0	0
	CDH74	9814	0	0	22	39	285	2115	3111	2735	1354	149	4	0
	CDH80	4055	0	0	1	3	59	910	1378	1175	512	17	0	0
(14) Wind	WSAvg	4.6	5.1	5.5	5.5	5.7	4.5	4.3	4.0	4.0	3.7	4.3	4.6	4.3
(15) Precipitation	PrecAvg	19.3	0.7	0.7	1.7	2.4	3.0	2.2	1.9	1.6	1.8	1.3	1.2	0.7
	PrecMax	29.4	2.2	2.4	5.4	6.0	9.6	6.0	5.2	5.6	4.9	5.4	3.5	2.2
	PrecMin	11.1	0.0	0.0	0.3	0.2	0.0	0.3	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.8	0.5	0.5	1.2	1.5	2.1	1.4	1.1	1.3	1.3	1.1	0.8	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	68.3	77.2	78.8	85.8	97.5	97.3	96.6	94.7	82.0	72.3	63.3
		MCWB	43.2	45.6	48.8	49.8	54.2	58.8	60.4	59.0	58.8	54.0	47.4	41.6
	2%	DB	59.2	62.7	71.9	73.2	81.2	92.8	94.6	91.5	89.9	78.9	67.7	58.8
		MCWB	40.2	42.3	46.1	48.9	53.7	58.8	60.4	58.5	57.4	52.3	46.2	40.4
	5%	DB	54.4	57.0	66.4	70.2	77.1	89.9	90.7	90.2	86.3	73.3	63.2	54.3
		MCWB	38.7	39.9	44.6	47.8	53.2	58.3	60.2	58.7	56.3	49.7	43.7	38.5
	10%	DB	51.6	52.4	62.7	65.7	72.4	85.9	88.1	87.7	81.9	70.0	59.1	50.2
		MCWB	37.3	37.8	42.4	46.5	52.3	57.6	60.3	58.8	55.6	49.0	41.5	36.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	46.2	49.7	52.2	60.6	67.0	66.0	65.4	62.2	56.1	48.6	42.7
		MCDB	62.3	66.7	75.6	70.0	73.4	81.4	78.5	76.3	75.5	74.8	70.4	61.0
	2%	WB	41.5	42.7	47.0	50.5	57.7	64.0	64.6	63.7	60.5	54.1	46.4	40.6
		MCDB	57.5	60.4	68.5	69.0	73.8	80.1	78.4	78.4	74.8	71.2	65.4	57.2
	5%	WB	39.2	40.6	44.9	49.1	55.5	61.8	63.8	62.3	59.2	52.3	44.4	38.9
		MCDB	53.5	56.4	64.2	66.9	69.4	77.9	78.2	78.2	75.3	69.5	61.5	53.6
	10%	WB	37.1	38.3	43.0	47.4	53.7	60.3	62.6	61.1	57.8	50.3	42.4	36.7
		MCDB	50.4	51.4	59.6	63.6	68.0	76.9	77.7	77.9	75.7	66.9	57.6	49.5
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	20.3	22.8	22.9	21.8	26.0	25.7	26.3	25.8	23.9	22.3	20.3
		MCDBR	27.4	26.5	29.6	30.2	29.7	31.8	30.1	30.7	32.1	32.4	29.6	25.4
		MCWBR	14.3	13.1	13.3	12.8	11.9	10.9	9.2	8.6	10.5	12.9	13.9	13.5
	5% WB	MCDBR	26.5	25.5	28.3	27.0	25.1	26.5	24.6	25.2	25.6	27.3	28.1	24.9
		MCWBR	14.6	13.2	13.1	11.9	11.7	11.1	9.0	8.5	9.6	11.1	13.6	13.8
(40) Clear-Sky Solar Irradiance	taub		0.244	0.253	0.278	0.306	0.326	0.333	0.365	0.356	0.313	0.279	0.258	0.246
	taud		2.577	2.545	2.499	2.464	2.474	2.468	2.423	2.435	2.541	2.612	2.590	2.577
	Ebn at Noon		303	313	313	308	301	298	287	288	296	299	294	293
	Edh at Noon		22	26	30	34	35	35	36	35	29	24	21	20
(44) All-Sky Solar Radiation	RadAvg		821	1144	1524	1761	1923	2228	2063	1835	1620	1194	854	702
	RadStd		37	63	122	120	156	119	94	79	96	78	52	44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page